Regime-Aware Nasdaq Allocation: Growth, Drawdowns, and Recovery

Yudian Pan

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Return observations through September 2026

Abstract

We examine whether monthly state-sensitive allocation improves the growth, drawdown, and recovery trade-off of a Nasdaq-oriented portfolio. A transparent rule and a constrained optimizer allocate across equities, bonds, gold, and reserves. Over January 2000–September 2026, their maximum month-end drawdowns are 17.23% and 27.94%, compared with 81.08% for Nasdaq buy-and-hold. Full-history protection exceeds that of constant portfolios with matched average holdings or realized volatility. The growth advantage is unstable across investment starting dates, however. During 2016–2025, both methods trail buy-and-hold, and a simple ten-month trend rule produces higher growth and smaller drawdowns than either. A loss classifier fails to improve on event frequency; authentic historical macroeconomic vintages add little forecast accuracy. Paired resampling does not establish positive growth outperformance. The results support a historical downside-control interpretation while exposing the opportunity cost and model dependence of additional complexity. Calculations use frozen monthly proxies: a daily-bar follow-up bounds one early-price discrepancy and tests next-session execution. Larger daily drawdowns and changed timing comparisons preclude interpreting monthly results as validated attainable returns.

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JEL classifications: G11, G17, C53, C61.

1 Introduction

An investor can recover a previous portfolio high while still lagging its purchasing power or a passive equity benchmark. These outcomes distinguish loss control from investment success. A Nasdaq-oriented allocation makes the distinction visible because its history includes both prolonged declines and extended periods of rapid equity appreciation.

This paper asks how observable market states and cost-aware allocation change that trade-off. Rule R combines equity trend indicators, defensive-asset filters, and state-dependent volatility budgets. Optimizer M maximizes a regularized forecast under risk, concentration, and trading-cost constraints. Both invest monthly without leverage across Nasdaq-100 equities, broad U.S. equities, aggregate bonds, gold, and reserves. The contribution is an empirical comparison of established methods, with recovery and information timing treated as separate outcomes and constraints.

The analysis distinguishes dynamic allocation from holding less equity. Constant portfolios match average holdings or validation-period volatility, and a ten-month trend rule supplies a simpler active comparator. Rolling investment starts assess dependence on the dot-com starting point. Forecast evaluation, macroeconomic vintages, and independent numerical accounting provide additional tests of the proposed interpretation. The unfavorable comparisons are central: the complex methods' full-history protection does not establish a stable growth advantage or a unique benefit from their state representation.

The study is a retrospective chronological evaluation. Earlier development already inspected the historical periods, including the later evaluation window. The current analysis retains the inherited R and M specifications and reports additional controls without choosing a replacement winner. All performance claims remain conditional on the documented monthly inputs and cost convention.

2 Related literature

Devanathan et al. (2026) study monthly unlevered stock, bond, gold, and cash portfolios over 2006–2026. They compare fixed allocations, volatility control, and convex optimization using simple and macroeconomic forecasts. Their study motivates the present Nasdaq-oriented extension. The asset universe, historical proxies, risk budgets, and accounting differ, so the present calculations are a separate implementation rather than a replication of their reported portfolios. Boyd et al. (2024) provide the broader cost-aware portfolio-construction framework.

Earlier research provides several models of state-dependent allocation. Guidolin and Timmermann (2007) study regime-switching asset-return distributions; Nystrup et al. (2019) combine state forecasts with multi-period portfolio decisions and drawdown control. Here, a deterministic indicator score defines operational allocation states. Moving-average timing also has a long history in tactical allocation (Faber, 2007). The simple trend comparator tests whether a more elaborate state rule adds useful performance in this particular universe.

The evidence on volatility management depends on design. Moreira and Muir (2017) report favorable risk-adjusted performance, while Cederburg et al. (2020) find that implementable out-of-sample benefits are less reliable across a wider strategy set. Bessler et al.

(2021) likewise show that allocation rankings vary across investment horizons and economic conditions in a different factor-and-sector universe. These results motivate comparisons across periods and baselines, without treating their samples as independent replications of the present study.

Selection and dependence constrain inference. Bailey et al. (2017) analyze backtest overfitting, and Lo (2002) show why Sharpe-ratio inference depends on the return process. This study reports paired block resampling and serial-dependence diagnostics. Their scope is the stated model family and observed history; unrecorded earlier specification searches remain outside the statistical correction.

3 Data, proxies, and the information clock

3.1 Asset universe and sample construction

The evaluation sample contains 321 monthly returns from January 2000 through September 2026. Nine preceding monthly returns, April–December 1999, are used for initialization. The starting wealth date is December 31, 1999. All base-case returns are in U.S. dollars and reinvest distributions. The model has no external cash flows or personal taxes.

Table 1: Research exposures and historical instruments

Exposure	Main instrument	Earlier observations and limitations
Nasdaq-100	QQQ	Actual ETF history from 1999; used as a proxy for a prospective QQQM holding, not as an identical fee history.
Broad U.S. equity	VTI	VTSMX before the VTI splice in 2001; mutual-fund and ETF histories are not identical trading vehicles.
Aggregate bonds	BND	VBMFX before the BND splice in 2007; includes non-Treasury risks and is not cash.
Gold	GLD	Indicative spot prices with an assumed custody charge before the 2004 splice; used as a proxy for prospective GLDM exposure.
Short reserve	Bill-rate proxy	Previous-month TB3MS divided by 1,200, less 9 basis points per year; not an observed SGOV total-return series.
Secondary benchmark	SPY	Actual S&P 500 ETF total-return proxy.

Notes: Exact monthly mappings and the frozen observations are supplied in `raw/`. An instrument's modern ticker does not imply that it existed throughout the sample.

The inherited prices originate from secondary-vendor adjusted-price tables (Digrin, 2026); the early gold segment uses an indicative spot source (MetalflowAPI, 2026). A documented 1999 warm-up supplements the evaluation history. Most price observations have only two decimal places. Rounding can matter materially for low-price bond-fund histories.

These are not independently audited exchange or custodian NAV records. Appendix E reports 39 independent aggregate issuer and SEC checkpoints. Modern comparisons differ by up to 13.72 basis points; an early QQQ market-return comparison differs by 77.31 basis points. NAV comparisons involve a different return basis. A daily-price audit narrows the early discrepancy, and fixed return perturbations and a complete QQQ source substitution preserve the central interpretation (Appendix E). No aggregate adjustment is imposed on the frozen observations.

The QQQ and GLD histories already reflect their historical fund-level expenses. We do not subtract today's QQQM or GLDM fee again. Conversely, the cash and pre-fund gold series include explicit modeling assumptions rather than observed ETF expenses. The share-class and tax consequences of replacing these instruments in a real account are not estimated.

3.2 Decision and observation timing

Let t denote a completed month-end and $r_{i,t+1}$ the subsequent monthly return. Indicators and forecasts use observations through t and determine weights applied to $r_{i,t+1}$. The inherited engine follows this information order but assumes a month-boundary fill. A separate daily-bar experiment freezes each target at the completed signal close and trades at the next observed opening bar in the 2012–2026 ETF era. The close and next-open branches use the same daily panel, including BIL reserves; their results are not spliced into the earlier monthly proxy history.

The initial covariance estimate uses only nine observations. Where a full lookback is unavailable, the implementation uses the available history; in particular, the early moving-average slope compares expanding rather than uniformly full ten-month averages. This initialization is disclosed rather than disguised as a long pre-2000 training sample.

For macroeconomic predictors, we acquire the Philadelphia Fed's official monthly vintage matrices for seasonally adjusted CPI and manufacturing production (Federal Reserve Bank of Philadelphia, 2026c,a). At signal month t, the feature constructor admits a vintage dated no later than $t-1$ and uses reference observation $t-2$. This extra vintage-month lag conservatively accommodates within-month release timing. Each growth ratio uses exact lags within one vintage. Appendix C specifies missingness, fallback rules, and the matched latest-vintage comparison.

The production definitions require particular care. The Philadelphia Fed's IPM history uses the Standard Industrial Classification (SIC) manufacturing series, whereas the inherited FRED IPMAN observations use the North American Industry Classification System (NAICS). Their differences combine definition and revision effects (Federal Reserve Bank of Philadelphia, 2026b; Board of Governors of the Federal Reserve System, 2026). The new comparison uses SIC on both sides to isolate the effect of vintage information within that definition. The inherited macro branch remains a separately labeled exploratory output.

Latest-vintage CPI serves only to evaluate ex-post purchasing power. Its overlapping nonmissing levels agree exactly with the inherited CPI observations. The real-performance sample ends in December 2025; the absent October 2025 observation remains missing, with elapsed calendar time preserved. The reserve sleeve uses a lagged bill-rate approximation, not an observed bill-fund total return or a complete historical release-vintage record.

4 Portfolio methods

4.1 R: transparent state-dependent risk allocation

For each equity exposure $j \in \{Q, U\}$, define a total-return price index $P_{j,t}$ and three binary indicators:

$$T_{j,t} = \mathbf{1}\{P_{j,t} > MA_{10,j,t}\}, \tag{1}$$

$$A_{j,t} = \mathbf{1}\{R^{(6)}_{j,t} > R^{(6)}_{c,t}\}, \tag{2}$$

$$S_{j,t} = \mathbf{1}\{MA_{10,j,t} > MA_{10,j,t-3}\}. \tag{3}$$

Here $R^{(6)}$ is the compounded return over the same trailing six-month window, including the reserve comparison. The state score is

$$z_t = \sum_{j \in \{Q,U\}} (T_{j,t} + A_{j,t} + S_{j,t}). \tag{4}$$

Scores of 5–6 designate risk-on, 2–4 transition, and 0–1 defensive. These are operational labels. Overlapping equity holdings and similar lookbacks make the six indicators dependent; a score of five is not five independent pieces of evidence or an 83% probability of a positive return.

Table 2: R allocation templates before asset filters and risk scaling

State	Nasdaq	U.S.	Bonds	Gold	Reserve	Risk budget
Risk-on	65%	20%	10%	5%	0%	15%
Transition	30%	15%	20%	15%	20%	12%
Defensive	10%	0%	35%	25%	30%	8%

Notes: Templates are stipulated policy assumptions, not optimized ex post weights. Risk budgets refer to estimated annualized volatility, not realized loss limits.

Bonds and gold independently must satisfy both their own ten-month trend test and six-month return test relative to reserves. A failed sleeve is moved to reserves rather than reassigned to another declining risky asset. Let v_t be the resulting four-dimensional risky

weight vector, and let $\widehat{\Sigma}_t$ be the monthly covariance estimate. Risk scaling gives

$$a_t = \min\left\{1, \frac{\sigma_{s_t}}{\sqrt{12 v_t^\top \widehat{\Sigma}_t v_t}}\right\}, \tag{5}$$

$$w_t = a_t v_t, \qquad w_{c,t} = 1 - \mathbf{1}^\top w_t. \tag{6}$$

The rule allocates the full portfolio between risky assets and reserves. Defensive templates retain limited equity exposure, and the estimated volatility budget determines the final scale.

4.2 M: simple cost-aware constrained optimization

M uses the preceding 36 months of excess returns when available, with a 12-month exponential half-life, and shrinks the estimated monthly mean halfway toward a stipulated prior:

$$\hat{\mu}_t = \mathrm{clip}\left(0.5\ \mathrm{EWMA}_{12}(r - r_c)_t + 0.5\mu_0, -0.025, 0.025\right), \tag{7}$$

where $12\mu_0 = (0.06, 0.045, 0.01, 0.005)^\top$. The monthly clipping limits and stipulated priors regularize the forecast.

Let $\widetilde{w} = (w, 1 - \mathbf{1}^\top w)$ contain reserve weight and let $\widetilde{w}_t^-$ denote drifted pre-trade weights. The general optimization family solves

$$\max_{w} \quad \hat{\mu}_t^\top w - \kappa \|U_t w\|_2 - c\|\widetilde{w} - \widetilde{w}_t^-\|_1, \tag{8}$$

subject to

$$w \geq 0, \qquad \mathbf{1}^\top w \leq 1, \qquad \sqrt{12 w^\top \widehat{\Sigma}_t w} \leq \sigma_t, \tag{9}$$

$$w_Q \leq 0.85, \quad w_U \leq 0.60, \quad w_B \leq 0.60, \quad w_G \leq 0.35, \tag{10}$$

$$w_Q + w_U \leq e_t, \tag{11}$$

$$\|w - (\mathbf{1}^\top w) b\|_1 \leq \mathbf{1}^\top w, \qquad b = (0.50, 0.20, 0.20, 0.10)^\top. \tag{12}$$

M sets $\kappa = 0$, $\sigma_t = 15\%$, and $e_t = 95\%$. The relative-weight constraint limits deviation from a reference allocation conditional on the total risky investment. It does not require maintaining a fixed amount of risk exposure.

For the robust variant, U_t is a diagonal scale for mean-estimation uncertainty. If the true mean lies in an ellipsoid $\{\hat{\mu}_t + U_t u : \|u\|_2 \leq \kappa\}$, its minimum portfolio mean is $\hat{\mu}_t^\top w - \kappa\|U_t^\top w\|_2$. Because U_t is diagonal, this gives the penalty in Equation (8). The uncertainty set remains an estimated modeling device, not a confidence guarantee.

The feasible set is convex and the objective is concave. The implementation uses an absolute-cost epigraph and SLSQP with analytic gradients, followed by a feasibility check.

Failed candidates fall back to reserves. For M, a separate HiGHS linear program bounds the objective from above using outer approximations of the quadratic risk constraint; feasible candidates supply lower bounds. All 321 monthly problems meet the stated numerical tolerance (Appendix A.3). This check establishes solution quality conditional on the fitted estimates.

4.3 Additional variants and forecasting experiment

MR replaces the simple mean with a regularized regression forecast. Training labels are three-month forward excess compound returns divided by three; all three months must have matured at the signal date. At least 60 valid labels are required, with at most about 120 recent labels, training-only standardization, and a ridge penalty of ten. Forecasts are shrunk toward the same prior and clipped. Appendix A provides the implementable details.

RM combines the regression forecast with $\kappa = 0.25$, the three state-dependent volatility budgets, and equity caps of 95%, 65%, and 25%. PM instead uses a logistic forecast p_t to set $\sigma_t = 15\%(1 - 0.5p_t)$ and $e_t = 95\% - 0.7p_t$. The macro extensions add three CPI and manufacturing features using either conservative as-of or latest-vintage observations. The extensions isolate the contribution of each additional modeling choice.

The classifier predicts the specific event

$$Y_t = \mathbf{1}\left\{\prod_{k=1}^{3}(1 + r_{Q,t+k}) - 1 \leq -0.10\right\}. \tag{13}$$

This event targets a specific three-month equity loss. A logistic model with four predetermined price features and an L_2 penalty is evaluated against a smoothed training-window event frequency. Forecast accuracy is evaluated separately from the resulting investment performance across all eligible dates.

5 Trading, costs, and evaluation

5.1 Self-financing accounting

Let h_i^- denote pre-trade dollar holdings and V^- their sum. A rebalance finds post-cost wealth V^+ and holdings $h_i^+ = V^+\widetilde{w}_i$ satisfying

$$V^+ = V^- - \sum_i c_i|h_i^+ - h_i^-| - fN_{\text{orders}} - C_{\text{FX}}. \tag{14}$$

Subsequent wealth is $V_{t+1}^- = \sum_i h_i^+(1 + r_{i,t+1})$. The base engine solves the self-consistency relation by fixed-point iteration; a second holdings implementation uses bisection. Initial

deployment is charged once as purchases from external cash. The reserve is a modeled investment sleeve, so switches to and from it are included in later turnover.

The base case sets $c_i = 10$ basis points per absolute dollar bought or sold. A complete switch from one asset to another can therefore incur both sale and purchase charges. Fixed dollar fees and initial/final currency conversion are separate scenarios. Base terminal wealth remains invested; final liquidation is not silently charged to active strategies alone. Fund-level costs already embodied in observed returns are not deducted again.

A two-percentage-point no-trade band applies only when the state has not changed and the current portfolio still meets the estimated risk and equity-cap checks. Price drift is retained between trades. Minimum-ticket fees, trading spreads, taxes, order fills, and whole-share constraints are distinct implementation issues; a monthly decision frequency does not imply only twelve orders per year.

5.2 Performance and recovery metrics

For N months, base CAGR is $(V_N/V_0)^{12/N} - 1$. The conventional reported volatility is $\sqrt{12}\,\mathrm{Std}(r_p)$ and the reported Sharpe ratio is $\sqrt{12}\,\mathrm{Mean}(r_p - r_c)/\,\mathrm{Std}(r_p - r_c)$. These are standardized monthly-sample summaries, not exact time-aggregation identities under serial dependence. Appendix D provides a separate finite-horizon diagnostic motivated by Lo (2002).

Define month-end drawdown as

$$D_t = 1 - \frac{V_t}{\max_{s \le t} V_s}. \tag{15}$$

Maximum drawdown is the greatest observed D_t. Time under water starts at a prior wealth high and ends when that high is regained. Durations count elapsed monthly steps through the recovery endpoint. An unfinished episode is right-censored. This measure differs from trough-to-recovery time. The latter ignores how long the decline itself took.

Relative downside is also evaluated over common Nasdaq-high-to-subsequent-month windows. We compare both investments over the same dates, rather than subtracting maximum drawdowns attained on different dates. These common-date losses measure benchmark-relative downside separately from each portfolio's own maximum drawdown.

Real wealth is $V_t^{\mathrm{real}} = V_t\,\mathrm{CPI}_0/\mathrm{CPI}_t$. Nominal recovery need not restore purchasing power. The calculation preserves unitized wealth without external contributions.

6 Research design and selection discipline

The study uses 2000–2009 as development history, 2010–2015 for one explicit method-selection exercise, 2016–2025 for the historical chronological audit, and January–September

2026 as a supplement. A 2012-onward comparison reduces reliance on pre-inception asset proxies. All dates and candidates remain visible.

Validation eligibility requires observed maximum drawdown no larger than 30%, annual two-way turnover no larger than six times wealth, and longest water duration no longer than 48 months. Among eligible methods the stipulated score is $\log(1 + \mathrm{CAGR}) - \frac{1}{2}\sigma^2$. This is a preference score, not a universal welfare criterion. It selects M; its name is then retained for the later audit regardless of its subsequent rank.

Each subperiod metric table starts with fresh cash and charges initial deployment, while estimators retain legitimately earlier observations. The no-trade state is reinitialized. In contrast, crisis-window returns and the serial-correlation diagnostic slice the continuous full-history portfolio. These accounting conventions are intentionally distinguished; the later audit is not presented as a single uninterrupted investor account that began in 2000.

Parameter choices were locally frozen for the preceding computational round, after earlier market histories had already been inspected. The 2026 data supplement is not an untouched prospective test. Sensitivity analysis characterizes nearby assumptions; it is not permission to replace a prespecified method with whichever variant wins after inspection.

6.1 Simple controls and attribution

For each active policy, the first constant control holds its arithmetic mean monthly post-trade weights. The second holds a fixed QQQ/reserve mixture whose realized volatility matches the active policy during calibration. For the 2016–2025 evaluation, both controls freeze their weights using only 2010–2015 observations, then rebalance monthly with the same cost accounting. Evaluation-period risk and exposure can differ from their calibration targets. Separate full-history matches use hindsight and serve only as descriptive attribution checks.

The trend comparator holds QQQ when its observed total-return index exceeds its trailing ten-month average, and reserves otherwise. It uses the same monthly clock and ten-basis-point trading charge. Its lookback is fixed before this revision's new comparisons; the historical sample remains previously observed. No additional predictor or asset is introduced.

7 Empirical results

7.1 Protection across the full history, slower growth later

R earned 9.17% annually with a 17.23% maximum drawdown; M earned 11.07% with a 27.94% drawdown. QQQ earned 8.82% with a 81.08% drawdown. The wealth and drawdown paths show how avoiding part of the prolonged early losses changed subsequent

Table 3: Full-history performance, January 2000–September 2026

Method	CAGR (%)	Vol. (%)	Sharpe	MDD (%)	Water
QQQ	8.82	22.89	0.41	81.08	175
SPY	8.23	15.10	0.48	50.79	75
Static 70/20/10	8.81	16.23	0.49	63.14	131
Volatility control	10.15	14.95	0.60	37.05	80
R	9.17	10.21	0.73	17.23	43
M	11.07	13.39	0.71	27.94	48
Trend (10 months)	9.44	14.15	0.58	36.52	86

Notes: 321 monthly returns. All methods use modeled costs and the same frozen dollar-return inputs. MDD is the positive magnitude of maximum month-end drawdown. Water is the longest elapsed time from a previous high through recovery, in months. Static 70/20/10 denotes Nasdaq/bonds/gold.

compounding (Figures 1 and 2). R's lower loss depth came with lower growth than M and higher trading intensity.

Table 4: Later historical evaluation, January 2016–December 2025

Method	CAGR (%)	Vol. (%)	Sharpe	MDD (%)	Water
QQQ	19.44	18.39	0.95	32.58	24
SPY	14.70	15.09	0.85	23.93	24
Static 70/20/10	15.75	13.59	1.00	26.74	24
Volatility control	15.34	14.76	0.91	25.52	26
R	13.09	10.87	1.00	13.91	19
M	14.83	14.66	0.88	27.94	29
Trend (10 months)	16.83	14.17	1.03	13.06	15

Notes: 120 months; fresh initial cash positions, with preceding observations available to estimators. This period was inspected in earlier development. Trend denotes the fixed ten-month QQQ/reserve rule.

The later period favored sustained equity participation. R and M earned 13.09% and 14.83%, against QQQ's 19.44%. R reduced the longest nominal water period to 19 months; M's 29 months exceeded QQQ's 24 months. Smaller maximum drawdown therefore did not guarantee faster recovery for M.

The simple trend rule earned 16.83% with a 13.06% maximum drawdown and 15 months under water. It exceeded both R and M on these three measures. R retained lower realized volatility, 10.87% versus the trend rule's 14.17%. Thus the comparison supports different risk preferences, but provides no general performance justification for the additional state-rule complexity in the later window.

7.2 How much comes from lower exposure?

In the full-history hindsight diagnostic, R's mean-holdings control had a 52.79% maximum drawdown, compared with R's 17.23%. M's control had a 59.86% drawdown, compared

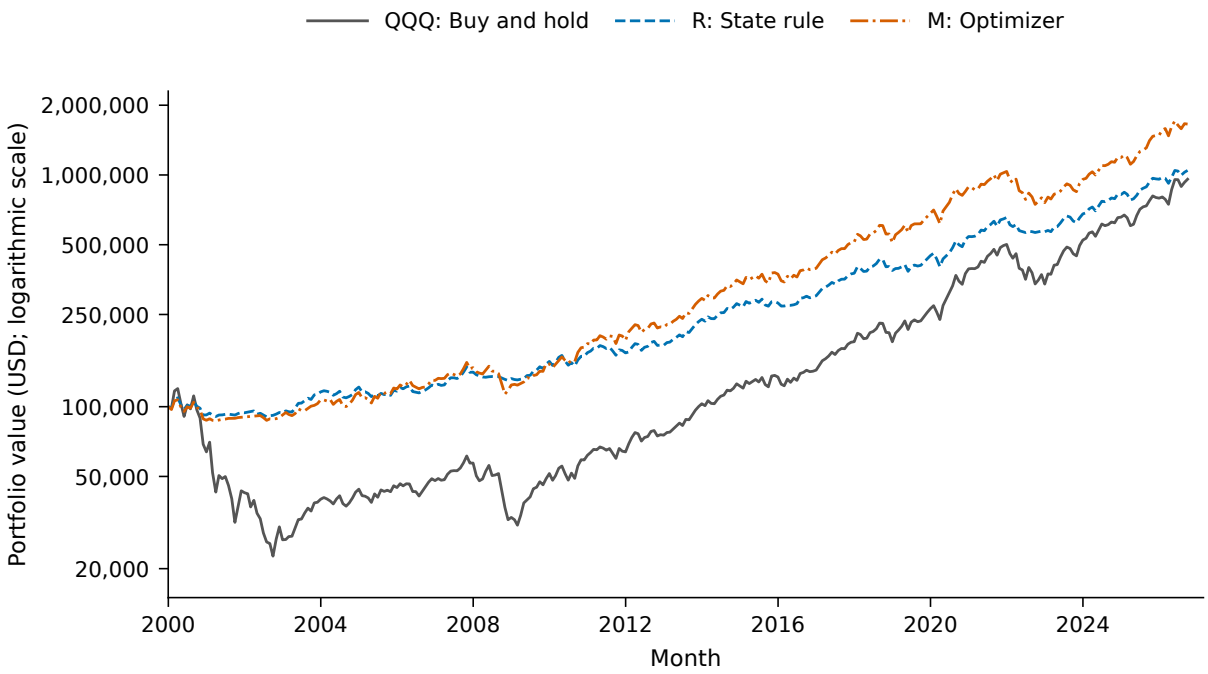


Figure 1: Wealth from $100,000 over January 2000–September 2026. The logarithmic axis shows all 321 modeled monthly returns for QQQ, R, and M, plus the initial valuation. Straight segments connect observations; they do not represent intramonth prices. Source data reproduce each plotted value.

Table 5: Constant controls calibrated in 2010–2015, evaluated in 2016–2025

Policy/control	CAGR (%)	Vol. (%)	MDD (%)	Water	Equity (%)
R	13.09	10.87	13.91	19	67.69
R mean holdings	14.76	13.40	24.96	24	75.41
R volatility match	13.86	12.38	22.45	19	67.25
M	14.83	14.66	27.94	29	81.42
M mean holdings	17.89	15.24	28.32	23	82.09
M volatility match	16.57	15.28	27.40	23	83.05

Notes: Mean-holdings controls fix each active policy's average five-asset weights. Volatility-match controls fix QQQ/reserve weights that match calibration-period volatility. Equity is average realized QQQ plus broad-equity weight during evaluation. Controls rebalance monthly; their evaluation risk need not equal that of the active policy.

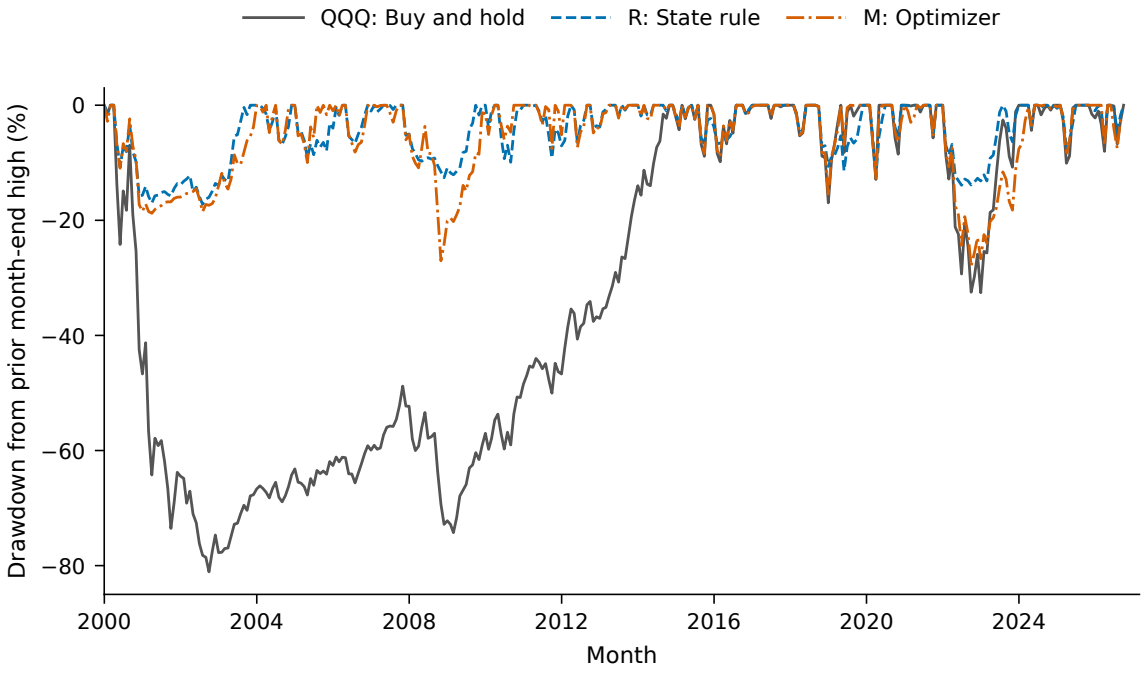


Figure 2: Losses from previous month-end wealth highs on the same full-history paths. Each curve includes all 322 valuations. These realized monthly paths have no sampling error bars; they omit intramonth drawdowns and next-session execution effects.

with M's 27.94%. The controls hold the same average asset vector, so lower average equity exposure alone does not explain the observed full-history protection. Volatility-matched QQQ/reserve portfolios also suffered larger drawdowns, 46.53% and 58.10%, respectively.

The validation-frozen controls reveal the later growth cost (Table 5). Both active policies trailed their mean-holdings and volatility-match controls in CAGR. R retained a smaller drawdown than either of its controls. M's drawdown was slightly smaller than its mean-holdings control but slightly larger than its volatility-match control. These comparisons identify a path-dependent downside benefit, not a stable improvement in every performance dimension.

A sequential component analysis gives the same qualified interpretation. Adding volatility scaling to R's fixed risk-on template reduced full-history drawdown from 67.34% to 39.63%. Adding states reduced it to 18.97%; the defensive-asset filters reduced it to 17.23%. The last step lowered growth from 9.81% to 9.17% and lengthened recovery from 41 to 43 months. Appendix B.1 retains both periods and the order-dependent nature of this attribution.

Table 6: Cumulative crisis-window returns

Method	2000–2002	2008	2020	2022
QQQ buy-and-hold	-73.36	-41.72	+48.63	-32.58
R: state rule	-6.02	-6.21	+20.88	-13.17
M: simple optimizer	-7.91	-15.79	+29.53	-26.68
MR: ridge optimizer	-7.91	-18.18	+28.41	-25.50
RM: robust state	-4.85	-9.62	+22.01	-23.05

Notes: Slices of the continuous full-history portfolio. The 2000–2002 column compounds three calendar years; other columns cover one year. These are calendar-window returns rather than peak-to-trough losses.

7.3 Crises and investment starting dates

R and M reduced losses during the dot-com and 2008 windows, but participated less in the 2020 rebound. R began 2000 in its risk-on state; volatility scaling nevertheless limited initial equity exposure to about 57.9%. Some protection therefore preceded a change in its state label. Its subsequent reversals and delayed re-entry remain in the recorded path.

Across all completed January-start ten-year windows, R beat QQQ's CAGR in 3 of 17 windows and M in 8 of 17. Both had smaller maximum drawdowns in all 17. The corresponding five-year counts were 7 of 22 and 8 of 22 for growth, and 22 of 22 for drawdown. Figure 3 displays all ten-year growth differences. These overlapping windows describe start-date sensitivity; treating them as independent binomial trials would exaggerate precision.

7.4 Forecasts and historical macroeconomic information

The loss classifier did not improve on the contemporaneous smoothed event-frequency benchmark. Restricting every three-month outcome to 2016–2025 leaves 118 forecasts and seven event windows. Brier scores were 0.05871 for the model and 0.05855 for the baseline; AUC was 0.230. The original 120-origin window extended outcome follow-up into March 2026 and gave the same qualitative result (Appendix C). Overlapping rare events limit inference; post-hoc reversal of the score would constitute a new model.

Table 7: Macroeconomic vintage comparison under unchanged ridge rules

Forecast	Full CAGR (%)	Later CAGR (%)	Later MDD (%)	Water
Price-only MR	10.55	15.56	28.96	26
As-of macro (SIC)	10.91	15.64	28.65	26
Latest macro (SIC)	10.59	15.65	28.63	26

Notes: Full history is January 2000–September 2026; later history is January 2016–December 2025. As-of and latest branches use the same SIC manufacturing definition and the same availability rule determined by the as-of branch. Water refers to the later period.

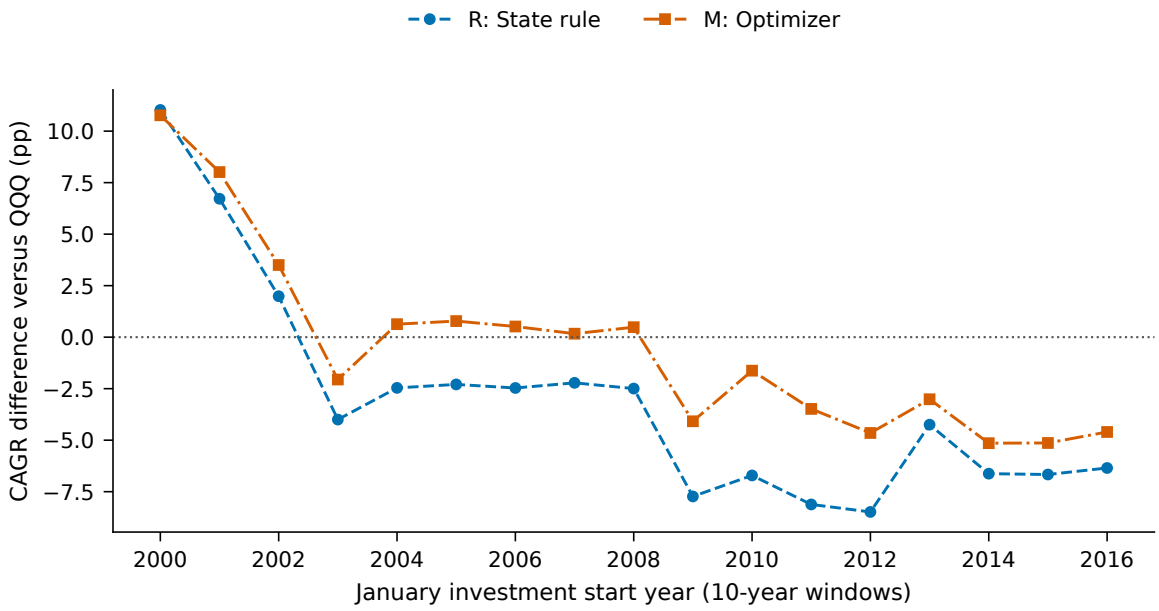


Figure 3: Start-date dependence of annualized growth relative to QQQ. Each point is a fresh-start ten-year investment, with historical estimator warm-up retained. All 17 January starts from 2000 through 2016 end by December 2025. Differences are percentage points of CAGR, not differences in mean log returns. Windows overlap.

The as-of macro extension increased full-history CAGR from price-only MR's 10.55% to 10.91%, but its later CAGR, 15.64%, was close to MR's 15.56%. The matched latest-vintage branch earned 15.65% later. Portfolio returns alone do not establish predictive value. Over 118 outcome-bounded origins and four risky assets, the as-of extension reduced mean squared forecast error by only 0.28%. A paired twelve-month-block interval for the loss difference crossed zero (Appendix C). These results provide limited support for adding these macro features under this specification.

7.5 Purchasing power and tail risk

Latest official CPI extends QQQ's longest water period to 206 months, compared with 175 months nominally. R's longest real recovery was 86 months and M's 73, reversing their nominal ranking. Real CAGR over 2000–2025 was 6.36% for R and 8.16% for M, against 5.59% for QQQ. Inflation raises the recovery threshold, so growth following the trough matters alongside the initial loss. The nominal and real endpoints differ explicitly; Appendix B gives the complete real-performance table.

Full-history monthly expected shortfall at 95%, measured as the mean loss among the worst 5% of observed months, was 5.93% for R, 7.72% for M, and 15.09% for QQQ. Annual two-way turnover was 2.35 times wealth for R and 0.95 for M. Lower tail loss therefore

came with additional transactions. Common QQQ-high-to-subsequent-month windows also show that M could lose 10.10 percentage points more than QQQ over the same dates, despite its smaller own-path maximum drawdown.

8 Uncertainty, costs, and interpretation

Twenty thousand paired circular-block resamples evaluate annualized mean log-return differences using twelve-month blocks. Six dynamic comparisons across two windows form the inherited family of twelve tests. None establishes positive growth outperformance after Bonferroni correction. Six additional comparisons against the validation-frozen and trend controls likewise produce no adjusted positive-growth finding. The associated effect sizes and intervals remain visible in Appendix D. Six- and 24-month block sensitivities do not support a stable positive-growth conclusion for R or M.

A second exercise reconstructs price paths, indicators, covariance estimates, forecasts, and portfolio decisions over 100 paired block-resampled asset histories. Growth-difference ranges cross zero for both policies. Drawdown reduction is more consistent, while some reconstructed histories have longer recovery than QQQ. The small replication count yields coarse tails. This exercise is descriptive process sensitivity; it neither repeats all historical model selection nor generates independent crises.

Costs and timing affect the economic interpretation. The 62-case sensitivity family varies risk budgets, covariance windows, costs, no-trade bands, information delay, and reserve remuneration. An additional month of signal delay raises M's maximum drawdown to 34.44%. Removing realized reserve interest lowers full-history CAGR to 8.66% for R and 10.82% for M. A fixed-policy replay separates mechanical fee drag from cost-induced re-optimization; increasing fees monotonically lowers wealth when the trade policy is held fixed.

The central inference concerns conditional downside control. Dynamic allocation improved full-history losses beyond constant average-exposure controls, but simpler timing challenged the case for additional complexity. R and M also trade growth against volatility and recovery in different ways. An investor's preferred metric therefore determines the ranking. Figure 4 illustrates this with the full static-allocation grid: its hindsight growth winner places 60% in gold, above M's 35% cap. A retrospective frontier under different constraints cannot identify an ex-ante optimal allocation.

9 Limits and implications

The daily follow-up changes the strength of several comparisons. For 2016–2025 at a ten-basis-point trading allowance, next-open R and M earn 12.88% and 14.86% annually, compared with 19.68% for QQQ and 16.74% for Trend10. R's maximum month-end draw-

down is 13.53%, below Trend10's 15.66%: the simpler rule's three-metric superiority in the original monthly calculation does not survive this execution convention. R's daily-close drawdown is 25.61%, substantially greater than its month-end measure. Monthly protection must not be presented as an intramonth loss limit. Appendix E reports the full matched experiment.

Three limits remain. First, a contemporaneous SEC close and daily dividend reconstruction constrain but do not fully identify the 2009 issuer discrepancy. A complete QQQ substitute and 26 fixed perturbations preserve the principal downside interpretation; unrelated mutual-fund and spot-gold errors are outside that bound. Second, opening bars and theoretical ex-date total-return units do not establish fill size, spreads, dividend payment-date cash availability, settlement, or whole-share feasibility. Third, prior inspection of the historical windows prevents prospective validation. A prospective evaluation requires a fixed protocol and subsequent observations.

The vintage reconstruction resolves a narrower problem. It supplies authentic monthly snapshots for the CPI and SIC production experiment, with conservative availability and explicit missingness. It does not reconstruct the legacy NAICS series or prove exact release-time availability for every source. Adding credit spreads or further indicators without a new hypothesis and fixed evaluation protocol would enlarge the search space without resolving the present forecasting weakness.

These qualifications align the interpretation with the volatility-management literature. The study documents protection in a concentrated equity setting across several different declines, while exposing the cost of reduced participation and the instability of relative growth. Its contribution lies in the joint assessment of growth, loss depth, recovery, simple controls, and information timing. Numerical optimality of the fitted M problem is useful for reproducibility, but remains separate from the quality of the return forecast.

10 Conclusion

Monthly risk-sensitive allocation reduced prolonged losses in the frozen 2000–2026 history, beyond what constant average-exposure controls achieved. That protection did not yield a stable growth advantage. The later period and rolling starts favored equity participation, and a simple trend rule outperformed both principal methods on growth, drawdown, and recovery in the frozen monthly 2016–2025 calculation. Its drawdown advantage over R does not survive the next-open follow-up. Failed loss prediction and small macroeconomic forecast gains further constrain the case for complexity. Bounded price sensitivities preserve that interpretation, while the observed daily-bar study exposes additional timing and valuation-frequency risk. This is a working-paper result, not certification of an executable investment strategy.

References

David H. Bailey, Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. *The Journal of Computational Finance*, 2017. doi: 10.21314/JCF.2016.322. URL https://scholarworks.wmich.edu/math_pubs/42/.

Wolfgang Bessler, Georgi Taushanov, and Dominik Wolff. Factor investing and asset allocation strategies: A comparison of factor versus sector optimization. *Journal of Asset Management*, 22:488–506, 2021. doi: 10.1057/s41260-021-00225-1.

Board of Governors of the Federal Reserve System. Industrial production: Manufacturing (naics) (IPMAN). FRED, Federal Reserve Bank of St. Louis, 2026. URL https://fred.stlouisfed.org/series/IPMAN. Accessed October 13, 2026.

Stephen Boyd, Kasper Johansson, Ronald Kahn, Philipp Schiele, and Thomas Schmelzer. Markowitz portfolio construction at seventy. *The Journal of Portfolio Management*, 50 (8):117–160, 2024. URL https://web.stanford.edu/~boyd/papers/markowitz.html.

Scott Cederburg, Michael S. O'Doherty, Feifei Wang, and Xuemin Sterling Yan. On the performance of volatility-managed portfolios. *Journal of Financial Economics*, 138(1): 95–117, 2020. URL https://www.lehigh.edu/~xuy219/research/COWY.pdf.

Nikhil Devanathan, Alexandros E. Tzikas, and Stephen P. Boyd. Simple dynamic stock/bond/gold portfolios. arXiv:2609.07946v1, September 7, 2026. URL https://web.stanford.edu/~boyd/papers/stock_bond_gold_portfolios.html. Preprint; distinct from the independent monthly extension in this paper.

Digrin. Monthly adjusted-price tables for QQQ, SPY, VTI, VTSMX, BND, VBMFX, and GLD, 2026. URL https://www.digrin.com/stocks/detail/QQQ/price/. Secondary vendor; the specific frozen observations and source mappings are retained in raw/.

Mebane T. Faber. A quantitative approach to tactical asset allocation. *The Journal of Wealth Management*, 9(4):69–79, 2007. URL https://www.pm-research.com/content/iijwealthmgmt/9/4/69.

Federal Reserve Bank of Philadelphia. Real-time data set for macroeconomists: Industrial production, manufacturing (IPM), 2026a. URL https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/ipm. Official monthly vintage matrix retrieved October 13, 2026; source hash retained.

Federal Reserve Bank of Philadelphia. Real-time data set for macroeconomists: Industrial production documentation, 2026b. URL https://www.philadelphiafed.org/-/media/FRBP/Assets/Surveys-And-Data/real-time-data/data-files/documentation/doc_ip.pdf. Section VIII, SIC manufacturing and classification history; accessed October 13, 2026.

Federal Reserve Bank of Philadelphia. Real-time data set for macroeconomists: Consumer price index (PCPI), 2026c. URL https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/pcpi. Official monthly vintage matrix retrieved October 13,

2026; source hash retained.

Kenneth R. French. Data library: Transition to crsp ciz files, 2026. URL https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Massimo Guidolin and Allan Timmermann. Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control*, 31(11):3503–3544, 2007. doi: 10.1016/j.jedc.2006.12.004.

Invesco. Invesco QQQ ETF: Fact sheet, 2026. URL https://www.invesco.com/us-rest/contentdetail?contentId=841e411c-a1eb-4541-8cb8-0fa603abea81&dnsName=us. June 30, 2026; page 2 market-return checkpoints; retrieved October 13, 2026.

Andrew W. Lo. The statistics of sharpe ratios. *Financial Analysts Journal*, 58(4):36–52, 2002. doi: 10.2469/faj.v58.n4.2453.

MetalflowAPI. Historical gold prices, 2026. URL https://metalflowapi.com/historical-gold-price/1999. Indicative spot-price proxy, not executable fund NAV.

Alan Moreira and Tyler Muir. Volatility-managed portfolios. *The Journal of Finance*, 72 (4):1611–1644, 2017. doi: 10.1111/jofi.12513.

Nasdaq-100 Trust, Series 1. Registration statement and prospectus, form 485bpos, 2007. URL https://www.sec.gov/Archives/edgar/data/1067839/000120677407000279/nasdaq_485bpos.htm. Page 6, annual NAV total returns for 2000–2006; accessed October 13, 2026.

Peter Nystrup, Stephen Boyd, Erik Lindström, and Henrik Madsen. Multi-period portfolio selection with drawdown control. *Annals of Operations Research*, 282:245–271, 2019. URL https://web.stanford.edu/~boyd/papers/multiperiod_portfolio_drawdown.html.

PowerShares QQQ Trust. Prospectus and financial statements, 2010. URL https://www.sec.gov/Archives/edgar/data/1067839/000110465910002985/a09-28465_1485bpos.htm. Consolidated closing prices and ex-dividend-date reinvestment specified on pp. 90–91.

PowerShares QQQ Trust, Series 1. Annual report for the year ended september 30, 2009, 2010. URL https://www.sec.gov/Archives/edgar/data/1067839/000110465910001821/a09-31668_2n30b2.htm. Pages 9 and 16; SEC Form N-30B-2; accessed October 13, 2026.

Patrick Schwarz, Dominik Walter, and Patrick Weiss. Rewriting CRSP's history: Impact of altered monthly returns on asset pricing. *Journal of Financial and Quantitative Analysis*, pages 1–37, 2026. doi: 10.1017/S0022109026102774. First View, published online March 24, 2026.

State Street Investment Management. SPDR Gold Shares: Fact sheet, 2026a. URL https://www.ssga.com/library-content/products/factsheets/etfs/us/factsheet-us-en-gld.pdf. June 30, 2026; market-value total-return checkpoints.

State Street Investment Management. State Street SPDR S&P 500 ETF Trust: Fact sheet, 2026b. URL https://www.ssga.com/library-content/products/factsheets/etfs/us/factsheet-us-en-spy.pdf. June 30, 2026; page 1 market-return checkpoints; retrieved

October 13, 2026.

U.S. Securities and Exchange Commission. Release no. 34-58943: Nasdaq omx phlx proposed rule change, 2008. URL https://www.sec.gov/rules/sro/phlx/2008/34-58943.pdf. Page 4 reports historical closing prices, including QQQQ at 38.91 on September 30, 2008.

Vanguard. Vanguard Total Bond Market ETF: Quarter-end fact sheet, 2026a. URL https://fund-docs.vanguard.com/F0928.pdf. June 30, 2026; market-price total-return checkpoints.

Vanguard. Vanguard Total Stock Market ETF: Quarter-end fact sheet, 2026b. URL https://fund-docs.vanguard.com/F0970.pdf. June 30, 2026; market-price total-return checkpoints.

Yahoo Finance. Historical daily bars, adjusted closes and corporate actions, 2026. URL https://finance.yahoo.com/quote/QQQ/history/. QQQ, VTI, BND, GLD, BIL and SPY; exact responses and hashes retained privately. Not exchange-certified auction records.

A Estimator and solver specification

A.1 Covariance and forecast details

Monthly risky excess returns are $x_s = r_s - r_{c,s}\mathbf{1}$. For the most recent $n = \min(36, \text{available})$ observations, normalized exponential weights have half-life six months. The weighted covariance uses the denominator $\max(1 - \sum_s a_s^2, 0.1)$. It is then shrunk 25% toward its diagonal and receives a 10^{-8} diagonal ridge. Base-case covariance uses at least the nine observed warm-up returns; alternative variance floors are separate sensitivity tests.

The simple mean uses the same observation window but a twelve-month half-life. Ridge forecasts use an intercept and 18 price features: for each risky asset, six- and twelve-month compounded returns, deviation from the ten-month mean, and the log ratio of recent three-month to twelve-month volatility; two additional features are the Nasdaq's trailing twelve-month drawdown and the annualized latest reserve return. Short histories use available observations. A minimum of 60 valid matured labels is required. Training weights have a sixty-month half-life; means and standard deviations are fitted inside the training window; feature standard deviations have a 10^{-4} floor.

For origins s with $s + 3 \le t$, the target is

$$y_{i,s} = \frac{1}{3}\left[\prod_{k=1}^{3}(1 + r_{i,s+k}) - \prod_{k=1}^{3}(1 + r_{c,s+k})\right]. \tag{16}$$

This arithmetic monthly-equivalent target is not a geometric excess-return identity. The normalized weighted least-squares objective has ridge penalty ten. The return prediction is shrunk 50% toward the stipulated prior. Uncertainty scales use weighted residual root mean squares divided by an approximate effective sample-size factor, with a floor of 0.00025 per month. The heuristic does not create a correctly calibrated confidence region for the mean.

PM's logistic features are Nasdaq's trend deviation, six-month momentum, short/long volatility log ratio, and trailing drawdown. The intercept is unpenalized, the other coefficients have an L_2 penalty, and the benchmark probability is $(n_{\text{events}} + 1)/(n_{\text{labels}} + 2)$. The training features are standardized using training-only values. PM falls back to its standard risk budget before a fitted probability is available.

A.2 Cost epigraph and no-trade behavior

The solver introduces five variables u_i satisfying $u \ge \widetilde{w} - \widetilde{w}^-$ and $u \ge -(\widetilde{w} - \widetilde{w}^-)$. Its objective adds $c\sum_i u_i$. The relative L_1 constraint is represented by all sixteen sign combinations over four risky weights. The annualized quadratic risk constraint is evaluated explicitly.

SLSQP uses an iteration limit of 160 and an objective tolerance of 10^{-10}. Failed or infeasible candidates are replaced by reserves. The no-trade risk check permits the stated numerical factor of 1.001 above the budget.

A.3 Independent objective bounds for M

For M, the cost epigraph makes the objective linear in (w, u); the only nonlinear constraint is the convex quadratic $q(w) = 12w^\top \widehat{\Sigma}_t w \le \sigma^2$. At any point x, convexity gives

$$q(w) \ge q(x) + \nabla q(x)^\top (w - x). \tag{17}$$

Consequently, each half-space

$$24x^\top \widehat{\Sigma}_t w \le \sigma^2 + 12x^\top \widehat{\Sigma}_t x \tag{18}$$

contains the feasible risk set. Retaining the linear portfolio constraints and replacing the risk set with these half-spaces yields a linear-programming outer approximation. Its maximized objective is an upper bound, U_t, while the objective of a feasible SLSQP candidate is a lower bound, L_t. The gap $U_t - L_t$ bounds candidate suboptimality for the fitted problem.

The audit uses HiGHS for the outer problem, with primal and dual feasibility tolerances of 10^{-9}. Initial cuts are placed at the feasible incumbent and at four axis points; additional cuts would be added at the LP solution if needed. It examines every monthly candidate, including months in which the no-trade band subsequently retains existing positions. All 321 problems satisfy the objective-gap tolerance 2×10^{-8}. The largest gap is approximately 6.1×10^{-10} in monthly-return objective units; the maximum measured primal violation is 7.3×10^{-15}. These are floating-point numerical bounds, not exact-arithmetic proofs. They concern the estimated one-period objective, independently of its ability to predict realized returns.

B Additional performance and attribution

Table 8: All inherited variants in the later evaluation

Method	CAGR (%)	Vol. (%)	Sharpe	MDD (%)	Water
QQQ	19.44	18.39	0.95	32.58	24
SPY	14.70	15.09	0.85	23.93	24
Static 70/20/10	15.75	13.59	1.00	26.74	24
Volatility control	15.34	14.76	0.91	25.52	26
R	13.09	10.87	1.00	13.91	19
M	14.83	14.66	0.88	27.94	29
MR	15.56	14.75	0.92	28.96	26
RM	13.82	12.91	0.91	25.03	26
PM	15.15	14.48	0.91	28.68	26
Legacy macro	15.65	14.72	0.93	28.63	26

Notes: Fresh starts over 2016–2025. MR uses ridge forecasts; RM adds state risk budgets and uncertainty penalties; PM uses loss probabilities. Legacy macro uses revised NAICS production with backward missing-value fallback and remains exploratory. Static 50/20/20/10 denotes Nasdaq/U.S. equities/bonds/gold.

Table 9: Live-ETF-era sensitivity, January 2012–September 2026

Method	CAGR (%)	Vol. (%)	Sharpe	MDD (%)	Water
QQQ buy-and-hold	20.16	17.43	1.06	32.58	24
SPY buy-and-hold	14.98	13.93	0.96	23.93	24
R: state rule	13.08	10.65	1.07	13.91	19
M: simple optimizer	15.66	14.18	0.99	27.94	29
MR: ridge optimizer	14.76	14.40	0.93	28.96	26
RM: robust state	12.93	12.61	0.91	25.03	26

Notes: ETF availability reduces reliance on pre-inception proxies. This remains a monthly-close model; reserves remain rate-based.

B.1 Sequential component analysis

The fixed risk-on template holds $[0.65, 0.20, 0.10, 0.05, 0]$. Scale only applies the 15% risk budget. States plus scale uses all three templates and budgets, and full R additionally applies the bond and gold filters. All variants use the same returns, covariance estimator, ten-basis-point charges, and two-percentage-point no-trade band. A separate holdings implementation reconstructs full R to within 10^{-9} in normalized wealth. The sequence defines the attribution order; interacting controls prevent a causal interpretation of individual differences.

Table 10: Real purchasing-power performance, January 2000–December 2025

Method	Real CAGR (%)	Real MDD (%)	Water (months)
QQQ	5.59	82.11	206
SPY	5.28	54.35	153
R	6.36	21.37	86
M	8.16	31.71	73
Volatility control	7.01	40.47	91

Notes: Latest official seasonally adjusted CPI; the missing October 2025 valuation is omitted without interpolation. Durations preserve elapsed calendar months. All listed longest episodes have recovered by the relevant endpoint.

Table 11: Tail risk and trading intensity over the full history

Method	Sortino	Monthly ES (%)	Two-way turnover	Orders/year
QQQ	0.58	15.09	0.04	0.04
SPY	0.70	9.43	0.04	0.04
Volatility control	0.94	8.23	0.70	13.01
R	1.16	5.93	2.35	30.65
M	1.14	7.72	0.95	24.37

Notes: Sortino uses the modeled reserve return as the monthly threshold and the root mean square of negative excess returns, including zero contributions from other months. The conventional annualization is $\sqrt{12}$. Monthly ES averages the worst $\lceil 0.05N \rceil$ months; it is an empirical summary, not a loss bound. Two-way turnover includes buys and sells.

Table 12: Investment-start sensitivity

Policy	Years	Windows	Growth wins	MDD wins	Median growth gap
R	5	22	7	22	-4.27
R	10	17	3	17	-4.00
M	5	22	8	22	-1.94
M	10	17	8	17	-1.63

Notes: Every completed January-start window ending by December 2025. Growth gap is the CAGR difference in percentage points. Counts refer to overlapping windows and have no independent-trial interpretation.

Table 13: R component analysis

	2000–2026			2016–2025		
Component sequence	CAGR	MDD	Water	CAGR	MDD	Water
Template	8.85	67.34	144	16.61	28.20	24
Scale only	9.54	39.63	80	15.12	25.10	25
States + scale	9.81	18.97	41	13.86	18.97	19
Full R	9.17	17.23	43	13.09	13.91	19

Notes: Percent units for CAGR and MDD; Water is elapsed months. Full history ends in September 2026, and the later period spans 2016–2025. These diagnostic variants do not replace R.

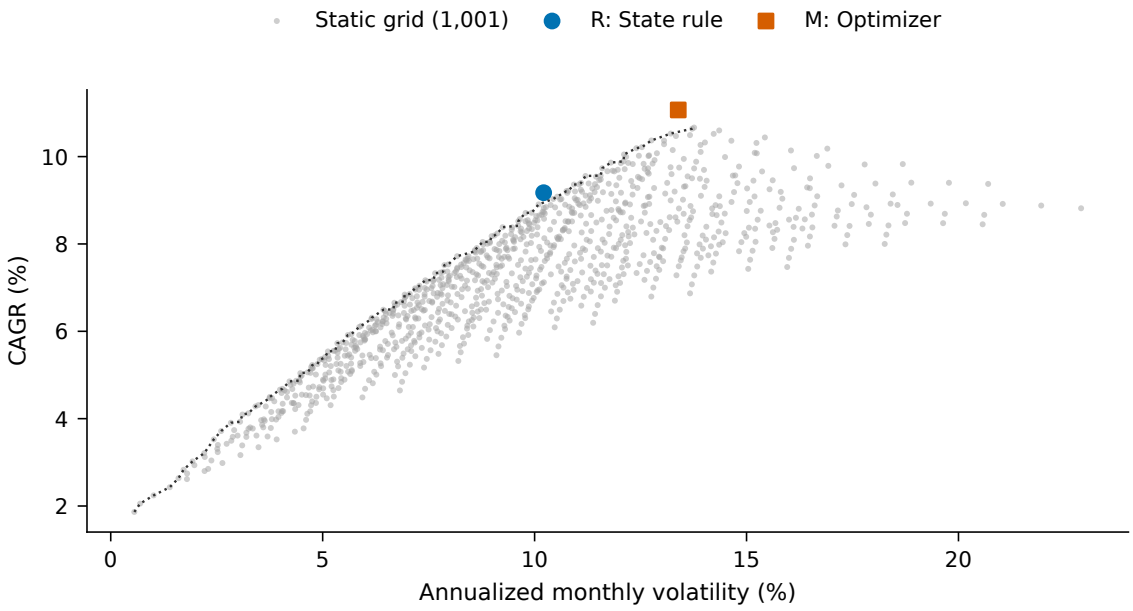


Figure 4: Descriptive full-history growth–volatility frontier. Every point in the 1,001-portfolio, ten-percentage-point static simplex enters the plot. The dotted boundary connects grid allocations not dominated on CAGR and volatility. R and M provide dynamic references. Static allocations rebalance quarterly and can hold 100% gold or reserves, unlike M's caps; the comparison uses full hindsight and different feasible sets. Recovery duration is a separate objective.

C Forecasts and macroeconomic vintages

The official PCPI matrix contains 335 monthly vintages from November 1998 to September 2026; IPM contains 767 from November 1962 to September 2026. The script reads unmodified workbook XML and retains the source hashes. At each signal month t, it chooses the latest vintage $v \leq t-1$ and reference observation $u = t-2$. The features are

$$\pi_t = C_u^{(v)} / C_{u-12}^{(v)} - 1, \tag{19}$$

$$\Delta_6 \pi_t = \pi_t - \left(C_{u-6}^{(v)} / C_{u-18}^{(v)} - 1 \right), \tag{20}$$

$$g_t = I_u^{(v)} / I_{u-12}^{(v)} - 1. \tag{21}$$

Each series uses its own admissible vintage; every numerator and denominator shares that series' vintage and index basis. Historical training features preserve the snapshot available at their own origin. The latest-vintage branch substitutes the September 2026 column while retaining the eligibility set determined solely by as-of information. It is a retrospective comparison, not an actionable forecast.

There are 318 complete macro feature sets among 321 actual allocation origins. Signals in November and December 2025 and June 2026 lack an exact required lag. Both branches

use the price-only estimator at those origins; training excludes incomplete macro rows. An additional feature set for the next decision is retained separately. No missing value is interpolated, carried backward to a different lag, or replaced with a future vintage in the as-of branch.

The Philadelphia Fed's documentation identifies SIC manufacturing, including the classification-history issues surrounding the 2002 benchmark revision (Federal Reserve Bank of Philadelphia, 2026b). FRED IPMAN identifies NAICS manufacturing (Board of Governors of the Federal Reserve System, 2026). Comparing their index levels does not identify a pure vintage effect. The same-definition paired branches in Table 7 avoid that conflation. The three predictor features leave the ridge penalty, prior shrinkage, caps, costs, and risk budgets unchanged.

Table 14: Outcome-horizon check for the loss classifier

Signal dates	Last outcome	N	Events	Brier	Base Brier	AUC
2016-01 to 2025-12	2026-03	120	7	0.05775	0.05762	0.230
2015-12 to 2025-09	2025-12	118	7	0.05871	0.05855	0.230

Notes: The first row reproduces the inherited origin-window definition and explicitly shows its 2026 follow-up. The second restricts outcomes to 2016–2025. The 118 origins run from December 2015 through September 2025. Event windows overlap; no probability threshold is tuned on either evaluation.

Table 15: Incremental macroeconomic forecast accuracy

Features	Price MSE	Macro MSE	Difference	95% range
asof	534.80	533.32	-1.48	[-4.40, +0.56]
latest	534.80	534.14	-0.67	[-3.58, +1.50]

Notes: MSE and differences are multiplied by 10^6. Each branch contains 118 origins and 472 asset forecasts of three-month compound excess return divided by three. Monthly four-asset loss vectors remain paired in 9,999 circular-block draws of length 12. The 95% percentile range is conditional on the realized fitted forecasts. This exploratory diagnostic was added after the portfolio specification and makes no multiplicity-adjusted discovery claim.

D Uncertainty and cost diagnostics

For each comparison, $d_t = \log(1 + r_{A,t}) - \log(1 + r_{B,t})$ and the statistic is $12\bar{d}$. Circular blocks preserve pairs, with 20,000 draws for the inherited twelve-comparison family. Centering the bootstrap statistic on its observed mean approximates the null for a one-sided positive-growth test. The six new control comparisons use 9,999 draws and their own Bonferroni family. These corrections concern the listed tests; they do not retrospectively preregister the study or correct unknown earlier searches.

Table 16: Conditional paired log-growth inference versus QQQ

Policy	Window	Log-growth gap	95% interval	Adjusted p
R	2000–2026	+0.33	[-6.34, +8.62]	1.00
M	2000–2026	+2.05	[-3.95, +9.65]	1.00
R	2016–2025	-5.46	[-10.70, +0.38]	1.00
M	2016–2025	-3.93	[-8.04, -0.34]	1.00

Notes: Log-growth differences and ordinary percentile intervals are in percentage points per year. The adjusted one-sided p values account for the complete inherited family of twelve tests. Familywise intervals and all candidates are retained in the CSV.

Table 17: Conditional inference versus the added simple controls, 2016–2025

Policy	Comparator	Log-growth gap	95% interval	Adjusted p
R	Mean holdings	-1.47	[-4.40, +1.65]	1.00
R	Volatility match	-0.68	[-3.75, +2.18]	1.00
R	Trend	-3.26	[-7.42, +1.34]	1.00
M	Mean holdings	-2.63	[-5.33, -0.29]	1.00
M	Volatility match	-1.50	[-4.62, +1.32]	1.00
M	Trend	-1.72	[-8.48, +4.84]	1.00

Notes: Units are annualized log-growth percentage points. Ordinary intervals describe effect uncertainty; adjusted one-sided p values test positive active-policy growth across six comparisons. A negative ordinary interval is not a multiplicity-adjusted inferiority finding.

Table 18: Paired process-reconstruction sensitivity

Policy	Paired metric	Median	Central 95% range
R	Growth difference	-1.80	[-8.37, +5.28]
R	Sharpe difference	+0.15	[-0.08, +0.35]
R	Drawdown reduction	+38.39	[+13.97, +61.26]
M	Growth difference	-0.56	[-5.14, +5.65]
M	Sharpe difference	+0.15	[-0.01, +0.40]
M	Drawdown reduction	+29.00	[+8.29, +54.32]

Notes: One hundred twelve-month-block histories, sampled jointly across the five assets including reserves. Each history reconstructs indicators and R/M decisions. Growth and drawdown differences are percentage points; Sharpe is unitless. Ranges are descriptive central 95% ranges with coarse tails, not calibrated future intervals.

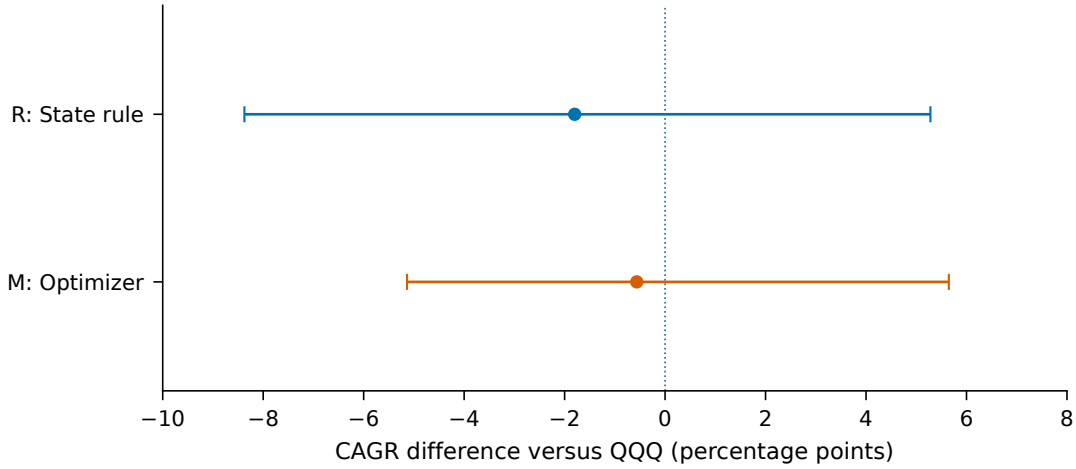


Figure 5: Growth differences under process reconstruction. Points mark medians and bars mark empirical 2.5th and 97.5th percentiles across 100 paired histories per policy. Each policy and QQQ face the same resampled asset history. All 200 policy–replication observations enter the calculation. Model-family selection is held fixed.

Table 19: Selected policy-recomputed sensitivities

Case	R CAGR (%)	R MDD (%)	M CAGR (%)	M MDD (%)
Base	9.17	17.23	11.07	27.94
Cost: 2.5 bp	9.37	16.94	10.58	28.63
Cost: 25 bp	8.79	17.81	10.75	29.32
Risk budget: 12%	8.49	13.45	10.74	24.90
Risk budget: 18%	9.49	21.56	11.73	29.65
One-month extra delay	9.22	20.27	11.10	34.44
Zero realized reserve yield	8.66	20.76	10.82	27.96
No no-trade band	9.15	17.45	11.08	27.94

Notes: Full-history results. Each case recomputes the allocation, so changes in optimizer cost assumptions can also change decisions. An extra month of delay is an adverse timing scenario, not a substitute for next-session opening data.

Table 20: Fee drag with the base policy held fixed

Per-side cost (bp)	R CAGR (%)	M CAGR (%)
0	9.43	11.17
2.5	9.37	11.15
10	9.17	11.07
25	8.79	10.91

Notes: Base trade dates and post-trade target weights remain fixed; a bisection holdings ledger recomputes dollar costs at each fee level. This is a counterfactual cost decomposition.

D.1 Serial dependence

For monthly excess return x_t, a separate stationary arithmetic-sum diagnostic estimates

$$\widehat{S}_{12} = \frac{12\overline{x}}{\sqrt{12\hat{\gamma}_0 + 2\sum_{k=1}^{11}(12-k)\hat{\gamma}_k}}, \qquad \hat{\gamma}_k = \frac{1}{N}\sum_{t=k+1}^{N}(x_t - \overline{x})(x_{t-k} - \overline{x}). \tag{22}$$

This concerns twelve-month sums, while main-table Sharpe ratios use conventional $\sqrt{12}$ scaling. Eleven sample autocovariances introduce estimation uncertainty, and stationarity is a substantive assumption (Lo, 2002).

Table 21: Conventional and serial-aware Sharpe summaries

Method	Full: conventional	Full: adjusted	Later: conv.	Later: adj.
QQQ buy-and-hold	0.41	0.37	0.95	1.05
R: state rule	0.73	0.74	1.00	1.10
M: simple optimizer	0.71	0.75	0.88	0.96

Notes: Later-period values slice continuous full-history policies; the main performance table restarts positions from cash. These conventions can produce small differences. No policy is reselected using this diagnostic.

E External evidence and execution boundary

E.1 Independent aggregate price checks

Twenty-nine modern issuer comparisons cover QQQ, SPY, VTI, BND, and GLD at June 2026 (Invesco, 2026; State Street Investment Management, 2026b; Vanguard, 2026b,a; State Street Investment Management, 2026a). The newly retrieved QQQ PDF has consistent June dates in rendered pages and extracted text. Published market returns differ from the compounded frozen inputs by up to 13.72 basis points. Ten earlier QQQ comparisons use SEC-hosted issuer filings (Nasdaq-100 Trust, Series 1, 2007; PowerShares QQQ Trust, Series 1, 2010). The twelve-month market-close return ending September 2009 differs by 77.31 basis points; the five-year annualized gap is 18.60 basis points. Eight NAV-basis comparisons differ by up to 80.81 basis points.

E.2 QQQ return conventions and bounded resolution

The fiscal-year discrepancy is not explained by citing a general data-vendor mechanism. Schwarz et al. (2026) document that the CRSP SIZ-to-CIZ transition changes monthly returns, largely through payout timing. Legacy SIZ monthly returns reinvest distributions at

Table 22: Aggregate issuer and SEC reconciliation

Series/basis	Windows	Max. absolute gap (bp)
QQQ early NAV	8	80.81
QQQ early market	2	77.31
BND modern market	6	2.96
GLD modern market	6	3.16
QQQ modern market	5	13.72
SPY modern market	6	12.42
VTI modern market	6	2.81

Notes: Absolute differences are basis points of the stated horizon return, annualized only beyond twelve months. Modern windows end June 2026. Early NAV comparisons cover calendar 2000–2006 and the year ending September 2008; early market comparisons end September 2009. Windows overlap. No statistical pass threshold or return correction is fitted.

month-end; CIZ compounds daily returns with reinvestment on ex-dates, as also described by French (2026). For a period without a split, the ex-date convention is

$$1 + R = \prod_{d=1}^{D} \frac{C_d + D_d}{C_{d-1}},$$

where D_d is the distribution per share on the ex-date. Corporate actions and units must be aligned before applying this identity. Our downloaded QQQ history has no split in the September 2008–September 2009 comparison window. These conventions do not imply that cash was available for reinvestment before its payment date.

A contemporaneous SEC exchange filing independently reports a September 30, 2008 QQQQ close of $38.91 (U.S. Securities and Exchange Commission, 2008, p. 4). The daily vendor reports the same opening endpoint and a September 30, 2009 close of $42.25. Their price-only return is 8.5839%, already above the issuer's 8.41% reported market total return. Positive dividend entitlements cannot lower terminal wealth below this price-only bound on identical endpoints, whether held as cash or reinvested at positive prices. The 2010 issuer prospectus specifies consolidated closing prices and ex-date reinvestment (PowerShares QQQ Trust, 2010, pp. 90–91); substituting today's midpoint convention for the historical definition would be unwarranted.

The adjusted-close factor A_d/C_d is retained separately from quoted OHLC. Floating-point display precision is immaterial at the reported scale. If the frozen adjusted endpoints 33.54 and 36.62 were rounded to the nearest cent, the total-return interval is [9.1519%, 9.2143%]. This can accommodate the small frozen-versus-daily difference, but not the issuer gap. Rounded dividends and the ex-date versus month-end choice likewise cannot explain approximately 75 basis points. Exact issuer quote inputs remain unavailable. An endpoint-field inconsistency or report error is possible; neither is established. The issuer total is not used to back-solve or overwrite any price.

Table 23: QQQ fiscal-year return reconciliation

Convention	Return (%)
Frozen adjusted monthly prices	9.1831
Daily adjusted closes	9.1608
Daily ex-date reinvestment	9.1608
Month-end reinvestment	9.1542
Unadjusted-for-dividend close ratio	8.5839
Issuer market total return	8.4100
Issuer NAV total return	8.4400

Notes: September 30, 2008 to September 30, 2009; returns in percent. Vendor dividend events are rounded to three decimals. The month-end alternative accrues dividends as receivables on ex-dates and reinvests at each month's final close. No payment dates are invented. NAV return is a separate economic basis.

A fixed follow-up plan replaces the complete QQQ monthly return history with daily adjusted-close ratios while retaining every other input. Separately, the twelve-month horizon return is shifted by ± 100 basis points. For each sign, the corresponding cumulative gross factor is either distributed evenly in log terms across twelve months or placed entirely in each of the twelve individual months. All 26 perturbations rerun QQQ, R and M without changing parameters. These are counterfactual sensitivity inputs, not recovered observations, and do not bound unrelated proxy errors.

Table 24: Full-history sensitivity to QQQ data uncertainty

Method	Inputs	CAGR	Max. DD
QQQ	Frozen	8.817	81.080
QQQ	Daily QQQ	8.826	81.078
QQQ	Fixed stress range	8.779–8.854	81.080–81.080
R	Frozen	9.173	17.229
R	Daily QQQ	9.180	17.041
R	Fixed stress range	9.163–9.183	17.229–17.229
M	Frozen	11.068	27.941
M	Daily QQQ	11.065	27.936
M	Fixed stress range	11.055–11.136	27.941–27.941

Notes: January 2000–September 2026; percent. Intervals are minima and maxima across the 26 fixed perturbations, not confidence intervals. The alternate feed can share an upstream lineage with the original vendor, so it is not treated as a wholly independent market record.

The original downside-versus-growth interpretation survives this finite audit. The precise cause of the issuer mismatch remains a documented limitation, rather than a reason to search indefinitely or to claim authenticated prices throughout the early proxy era.

E.3 Observed daily bars and next-session execution sensitivity

There is no universal academic convention that makes a reported opening bar a guaranteed fill. Monthly adjusted-close research with explicit proportional costs is common: Devanathan et al. (2026), for example, use daily adjusted closes and month-end rebalancing. That convention is a portfolio study, not opening-auction validation. We therefore retain the inherited monthly experiment and add a matched execution-clock sensitivity using actual daily bars (Yahoo Finance, 2026).

Each of QQQ, VTI, BND, GLD, BIL and SPY has all 3,708 expected common XNYS sessions from December 30, 2011 through September 30, 2026 in the downloaded panel. Dates, unique observations, positive prices, OHLC bounds and corporate-action records are checked. Actual ETF observations replace the four asset sleeves and BIL replaces modeled reserves. All 36-month estimation histories needed for R and M in the 2012-start experiment lie within observed ETF coverage. Earlier mutual-fund and indicative spot observations retain their monthly-only interpretation. No open is imputed for an unavailable instrument.

Let $C_{i,d}$ and $O_{i,d}$ be vendor historical close and open, and $A_{i,d}$ adjusted close. Define $F_{i,d} = A_{i,d}/C_{i,d}$ and $\widetilde{O}_{i,d} = O_{i,d}F_{i,d}$. The observed opening quote is preserved separately; $\widetilde{O}$ expresses it in total-return units. Targets, covariance, state and the no-trade decision are frozen at the previous completed month-end. Existing holdings bear $\widetilde{O}_{i,d}/A_{i,d-1}$ before the fee-consistent rebalance; new holdings then earn $A_{i,d}/\widetilde{O}_{i,d}$. Undeployed initial cash earns no overnight ETF return. Both branches use the same information set, prices, reinvestment convention and costs, isolating the decision-to-fill delay within this panel.

A second fixed cost of 25 basis points per dollar traded is reported above. Costs are stipulated aggregate allowances, not measured broker spreads or market impact. Four added checks independently reconcile the same-panel close branch to the original monthly engine, verify the no-cost buy-and-hold opening-price identity, perturb the next open without changing its already frozen target, and test self-financing fees. An observed bar is not proof of auction participation, available size or an executable limit order. The total-return-unit convention also does not model actual distribution payment dates, settlement, whole shares or taxes. Accordingly, this is an empirical execution sensitivity with a clear clock, not certification of a live investment strategy.

Table 25: Daily-panel timing sensitivity at 10 basis points per dollar traded

Period	Method	Clock	CAGR	Month DD	Daily DD
2012–2026	QQQ	Close	20.15	32.58	35.12
2012–2026	QQQ	Next open	20.00	32.58	35.12
2012–2026	R	Close	13.07	13.94	25.62
2012–2026	R	Next open	12.68	13.53	25.61
2012–2026	M	Close	15.61	27.92	29.48
2012–2026	M	Next open	15.40	27.77	29.32
2012–2026	Trend10	Close	15.43	19.78	28.56
2012–2026	Trend10	Next open	15.14	21.45	28.56
2016–2025	QQQ	Close	19.42	32.58	35.12
2016–2025	QQQ	Next open	19.68	32.58	35.12
2016–2025	R	Close	13.05	13.94	25.62
2016–2025	R	Next open	12.88	13.53	25.61
2016–2025	M	Close	14.76	27.92	29.48
2016–2025	M	Next open	14.86	27.77	29.32
2016–2025	Trend10	Close	16.77	13.09	28.56
2016–2025	Trend10	Next open	16.74	15.66	28.56

Notes: CAGR, month-end maximum drawdown and daily-close maximum drawdown, in percent. The boundary-close branch remains an idealization; the next-open branch uses observed opening bars. BIL, not the inherited bill-rate proxy, supplies reserve returns. The second period ends December 2025. All methods retain their specifications; the ten-month trend rule is the inherited simpler control.

Table 26: Next-open scenarios at 25 basis points per dollar traded

Period	Method	CAGR	Month DD	Daily DD
2012–2026	QQQ	19.99	32.58	35.12
2012–2026	R	12.31	13.84	25.61
2012–2026	M	16.07	26.73	28.46
2012–2026	Trend10	14.59	22.85	28.56
2016–2025	QQQ	19.66	32.58	35.12
2016–2025	R	12.48	13.84	25.61
2016–2025	M	15.95	26.73	28.46
2016–2025	Trend10	16.23	15.92	28.56

Notes: Percent. M anticipates each cost in its optimization objective, so this is a cost-aware policy scenario, not a fee-only deduction from fixed trades. Higher stipulated costs can alter allocation enough to raise realized returns; no scenario is selected as a replacement policy.

F Data, software, and declarations

Reproducibility. A clean Python environment recomputes the 66 inherited period–method summaries, 62 sensitivity cases, 1,001 static allocations, and 100 process-reconstruction histories. The greatest CAGR difference from the inherited core outputs is 1.64×10^{-12}; this agreement uses the same frozen inputs. The archive includes deterministic scripts, pinned dependencies, source hashes, exact figure data, and the canonical LaTeX manuscript. The documented runner distinguishes recomputation from checks of stored results. The finite follow-up adds 78 fixed-shock strategy runs, six source-comparison runs, 32 daily execution scenarios and four focused tests. Its public synthetic demo exercises the unchanged core; it does not reproduce empirical findings without authorized data.

Availability. The public package contains this paper, original code, schemas, source identifiers, access instructions and a fictional demonstration. Restricted histories, reconstructible datasets, correspondence and third-party full texts remain private. Empirical replication requires authorized data. This working paper is neither peer-reviewed nor a certified executable strategy.

AI disclosure. ChatGPT provided substantive research-design, source-search, coding, analysis, translation and writing assistance. One research agent used deterministic tests and selected separate implementations: computational self-review, not independent peer review. The human author approved the manuscript and accepts responsibility, without claiming personally to have written, rerun or reviewed every line of code.

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